6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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1 Introduction

Clouds are a major component in regulating the Earth's radiative budget. However, they exhibit the largest spatial and temporal variability compared to gases and aerosols (int). Additionally, the water, ice, and mixed-phase hydrometeors within clouds can vary significantly. Because of this, clouds remain poorly understood, particularly due to data gaps in regions sensitive to climate change variability, such as the Mediterranean basin. (Marinou et al.). In general, there are few studies on clouds using ground-based remote sensing measurements in Europe, with many located in Northern Europe (Stefan Kneifel et al.; Tatiana Nomokonova et al.; Bühl et al.; Peggy Achtert et al.). In the western Mediterranean basin, the most valuable database of cloud remote sensing measurements is provided by the AGORA ACTRIS CCRES station (Granada station), in the southeast of the Iberian Peninsula. Clouds in this region form through various mechanisms, including thermodynamic conditions, Sahara dust intrusions Casquero-Vera et al., and cut-off lows (COLs) systems, which create environments conducive to storms. These complex formation mechanisms pose significant challenges for accurately representing these clouds in atmospheric models. Consequently, there is a critical need for enhanced cloud analysis and data collection in the Mediterranean region.

Hence, in order to improve cloud analysis, the AGORA (Andalusian Global ObseRvatory of the Atmosphere) station has been part of Cloudnet since XXX and was integrated into ACTRIS CCRES (where ACTRIS stands for Aerosols, Clouds, and Tace gases Research InfraStructure, and CCRES for Centre for Cloud Remote Sensing) in 2023 as a new facility. The Cloudnet processing chain (J. A. et al.) performs extensive quality assurance on raw instrument for providing high-level products, such as target classification, liquid water path, and droplet effective radius. This chain standardizes data processing within the cloud remote sensing community. Their products are derived using a synergistic approach, integrating vertically pointing measurements from ground-based instruments like Doppler cloud radars, microwave radiometers, and ceilometers. Specifically, the target classification product, generated through these three instruments, has been utilized in numerous studies for cloud typing and multi-year analysis. For example, Stefan Kneifel et al. used this product in the German Alps to analyze ice clouds and precipitation variability at a high-altitude location. The authors found diurnal cycles in summer related to convective clouds, while other months are influenced by large-scale advection of clouds and precipitation. Similarly, Tatiana Nomokonova et al. investigated cloud profiles at the Ny-Ålesund station in the Arctic, noting a maximum cloud occurrence and higher cloud base in summer, differing from satellite observations. Multi-layer and single-layer clouds showed similar frequencies, with mixed-phase clouds being the most frequent single-layer cloud. These mixed-phase clouds exhibited higher thickness but lower ice water content compared to ice clouds. In contrast, Răzvan Pîrloagă et al. studied cloud profiling at the Bucharest–Magurele site, northeast of the Granada station, finding ice clouds to be the most frequent, unlike in Arctic studies. These studies rely on cloud classification methods that categorize clouds based on time or profile into different categories, e.g. liquid clouds, ice clouds and mixed-phase clouds. However, this approach can result in inhomogeneous cloud structures by classifying different cloud types within the same cloud, potentially skewing cloud property statistics and complicating further analysis.

This particular feature can produce high correlations between cloud types, significantly affecting cloud property statistics at some stations. This makes analyzing cloud properties challenging, as the method influences the values, which should not depend on the classification approach. Our study aims to improve cloud profiling and its physical properties using Cloudnet post-processed data by applying a novel cloud classification method. This new approach was motivated by Bühl et al., which deals with the inhomogeneity on cloud classification incorporating a minimum time between profiles for changing from one category to another (e.g. ice to mixed-phase), avoiding sharp changes of cloud types within the same cloud. Nevertheless, this method can also broke the cloud into different types depending on the time interval. Thus, we introduce a method based on identifying groups of connected hydrometeors, ensuring homogeneous classification and low correlation among cloud types. Therefore, we explore the advantageous to utilize this new approach in relation to the time-based classification. Furthermore, this method were used to evaluate the annual, seasonal, and daily cycles of cloud macrophysics (cloud base, top, and thickness) and microphysics (LWC, IWC, r_{liq}, and r_{ice}). These cloud properties were retrieved using Cloudnet algorithms based on A. S. Frisch et al.; Robin J. Hogan et al.; Hannes Griesche et al.; Julien Delanoë et al..

The experimental site, datasets, instruments and products are describe in Section 2. The proposed new approach on cloud classification algorithm and the cloud assesment are presented in Section 3. Section **??** analyses the annual, seasonal, and daily cycle of cloud types, as well as the base and top height, and thickness. In addition, this section discuss the liquid and ice properties variability for different clouds. Finally, Section **??** summarizes the main findings and discuss their importance to improve further cloud statistics analysis.

2 Experimental site and Instrumentation

An extensive dataset spanning the years 2018 to 2023 was utilized in this study. Specifically, high-level post-processed data from Cloudnet (**?**) such as categorize, target classification, and microphysical files, along with humidity and temperature profiles generated by the microwave radiometer, were employed. Figure 1 illustrates the monthly Cloudnet high level products available for the AGORA ACTRIS-CCRES station (37°2'E, 3°6'S), at 680 m of altitude. The availability of these data depends on simultaneous vertically pointing measurements of the microwave radiometer, ceilometer, Doppler cloud radar, as well as the meteorological model products - the ECMWF model (acronym for European Centre for Medium-Range Weather Forecasts). This figure shows that data collection efforts improved over time, with 2020 exhibiting the highest and most consistent data availability throughout the year. In contrast, 2018 had no data for the early months, indicating that data collection commenced mid-year. The white bars denote periods with missing data due to instrument maintenance, technical issues, and scanning measurements. Despite these challenges, these three instruments have been continuously operated since 2018.

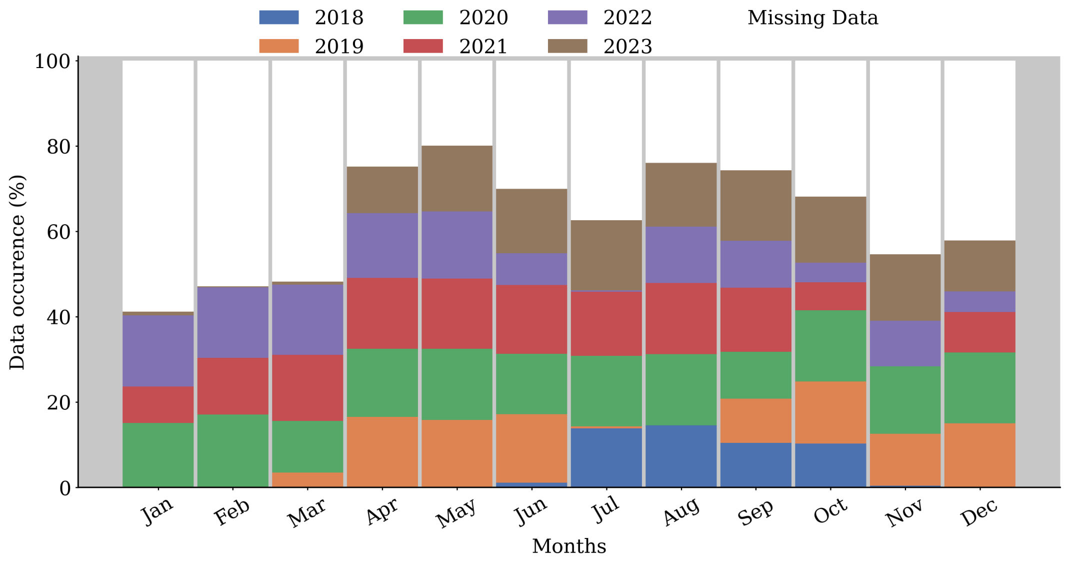


Figure 1. High level post-processed data availability per month at the AGORA ACTRIS CCRES station. This dataset represents more than half a decade of synergic ground-based measurements at this station. Each color represent the data availability for different years.

It is worthy to note that this dataset represents a robust database for cloud statistics, providing comprehensive data for cloud analysis over an extended period. Notably, it stands as the only cloud remote sensing dataset available in Spain, making it an invaluable resource for regional climate studies. The ACTRIS-CCRES Granada site is uniquely positioned as one of the few locations dedicated to cloud studies in a warm region that is frequently affected by synoptic conditions, Sahara dust intrusion, and thermodynamic conditions. Marinou et al.; F. J. Bello-Millán et al.; Casquero-Vera et al.. This dataset contributes significantly to filling the gap in cloud data for this climate variability hotspot, enhancing our understanding of cloud dynamics and aiding in the improvement of predictive models and climate studies in these critical areas.

2.1 Cloudnet database

2.1.1 Instruments

The Cloudnet target classification utilizes the remote sensing state-of-art instruments to study cloud properties and atmospheric phenomena. At Granada station, since 2010, a RPG-HATPRO G2 microwave radiometer has been employed, featuring 11 channels around the water vapor absorption band (23.8 GHz), 9 channels around the oxygen absorption band (52 GHz), and

10 channels around the liquid water absorption band (31.4 GHz). This instrument measures brightness temperature with a resolution of 0.3-0.4 K, which is subsequently used to calculate the liquid water path (LWP) and derive the liquid water content (LWC) for liquid clouds. Complementing this, since 2015, a CHM15k Nimbus ceilometer operating at a power of 8.4 μJ and wavelength of 1064 nm has been used to provide the attenuated backscattering coefficient of aerosols and cloud droplets. This instrument is used to estimating cloud base height and detecting aerosol layers due to the sensitivity of this wavelength to small particles such as aerosols and water droplets. Additionally, a dual polarimetric RPG-FMCW 94 GHz Doppler cloud radar were installed at the our station in 2018. It measures the Doppler velocity spectrum (DVS) for horizontal and vertical linear polarizations at 94 GHz. Unlike traditional single polarization measurements that retrieve only reflectivity (Z), mean Doppler velocity (ν_D), and spectral width (S_ω), this advanced radar also derives the linear depolarization ratio (LDR), enabling the detection of the melting layer, aerosols, and insects as described by Hogan and O'Connor. Table 1 provides a summary of the cited instruments, detailing their operational frequencies or wavelengths, the types of measurements they perform, their products, their measurement ranges, data collection intervals, and relevant references for a detailed description. The unique variables not processed by Cloudnet were humidity (q) and temperature (T) profiles, which were given by the microwave radiometer.

Table 1. Summary of instruments and their data products

Instrument	**Bands / wavelenghts**	**Measured**	**Products**	**Range**	**Time**	**References**
94 GHz Cloud Doppler Radar	94 GHz (W-band)	Doppler velocity spectrum (DVS)	ν_D , Z, S_w	[13,51] m[a]	[1.1, 3.6] s[b]	Nils Küchler et al.
MWR RPG	22-31 GHz (K-bad)	Brightness	q and T	[10, 300] m[c]	[2, 2.5] min	Francisco Navas-Guzmán et al.
HATPRO	51-58 GHz (V-band)	Temperature	LWP	-	2 s	
CHM15k Nimbus Ceilometer	1064 nm		β	15 m	15 s	A. Cazorla et al.

Note that the range and time resolution of the cloud radar can vary in altitude and time. It depends on the radar chirp table configuration, where the time resolution, the maximum unambiguous velocity (Nyquist velocity), and the maximum range are defined. These three variables are interdependent and can be strategically adjusted to meet specific objectives. In our case, this table was modified several times throughout the analyzed period, changing these resolutions in a complex way along the entire time series. In addition, range resolution of temperature and humidity profiles given by the microwave radiometer also can be modified, therefore, it also vary with time.

[a]This range vary with the height, and depends on the chirp table configuration

[b]It also depends on the chirp type

[c]Humidity and temperature profiles has a range varying from 10 to 150 m in the first 4.4 km and from 50 to 300m between 4.4 and 10 km

2.1.2 Data processing

Instrument datasets are processed by Cloudnet processing chain (J. A. et al.), which sampled data on a common grid with a time resolution of 30 s and height bins based on the cloud radar range resolution. Taking into account the variability of the radar range resolution with time, to ensure consistency in the vertical pointing measurements, a regridding process was implemented to facilitate monthly statistical analysis when needed. The regrinding involved interpolating measurements onto a height grid with a 30 m resolution and was applied to every inter-annual profiling. Additionally, Cloudnet performs corrections for reflectivity gas and liquid attenuation, along with error estimation and quality flag assignment. Specifically, liquid attenuation corrections on radar reflectivity are crucial during raining events, as rain droplets can attenuate a significant fraction of the radar signal.

The processed database are then used as input to generate the target classification product, which will be used to classify clouds into different categories. This product assigns an integer ranging from 0 to 10 at each pixel in the Cloudnet processed grid. Each number are related to atmospheric elements in the following order: clear sky (0), droplets (1), drizzle or rain (2), drizzle & droplets (3), ice (4), ice & droplets (5), melting ice (6), melting & droplets (7), aerosol (8), insects (9), aerosol & insect (10). These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature (T_w) profiles is used to determine the 0°C isotherm and establish warm and cold regions to identify liquid and ice hydrometeors. Further, liquid droplets boundaries were resolved with lidar attenuated backscatter coefficient (β), and radar reflectivity. The first is sensible to small liquid droplets present in cloud base, as the second are able to penetrate clouds and detect their tops. In addition, liquid layers respect a threshold of T < -40 °C, since it is not possible to keep water in liquid phase bellow that temperature. Moreover, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echos analysis once particles boundary were already determined. Therefore, falling cold particle were just classified as ice, without distinguishing ice from precipitating ice. Next, melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 s^{-1} within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on Hogan and O'Connor.

Despite the convenience of using Cloudnet products, target classification at the Granada showed instances of misclassification involving ice, melting, and rain pixels. Figure 2 illustrates a thinner and lower precipitable ice cloud present between 08:00 and 20:00. However, there is no evidence of cloud in the attenuated backscattering and radar reflectivity. These misclassifications stem from inaccurate representation of temperature profiles by models in this region, which subsequently affect cloud classification accuracy. To mitigate these issues and prevent misinterpretation of future cloud statistics, we refined the classification algorithm (see section 3) to better handle these cases.

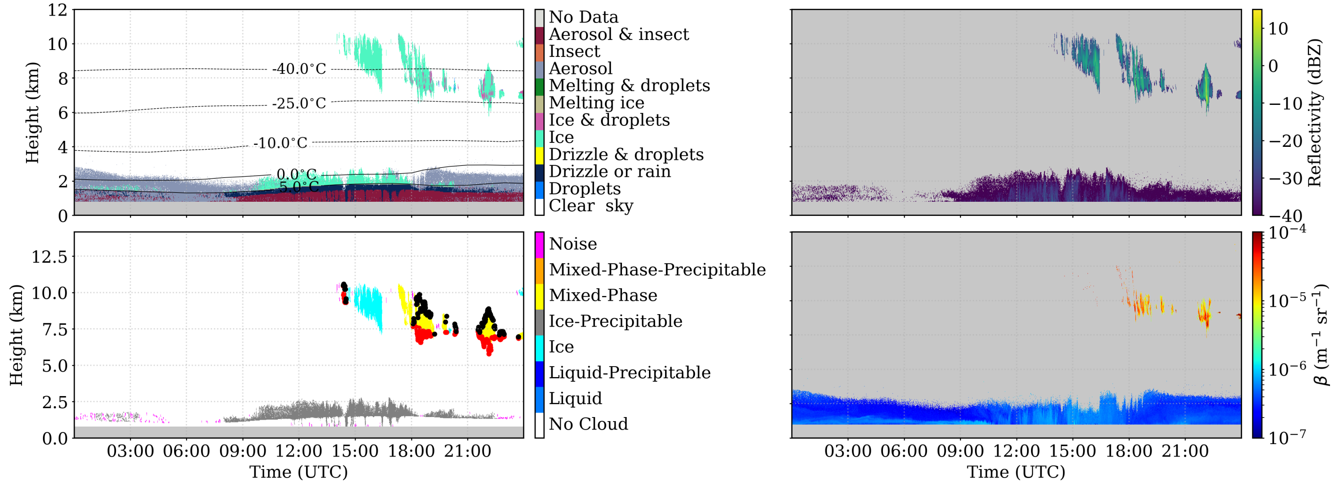


Figure 2. Temporary figure. A case study of Cloudnet hydrometeor misclassification. (a) Target classification product; (b) Radar reflectivity factor from Doppler cloud radar; (c) Attenuated backscattering coefficient from ceilometer. No evidences of clouds in the ABL can be identified by the instruments.

Also, specially important high level data such as liquid water content (LWC), ice water content (IWC), droplet effective radius and ice effective radius is described in table 2. For each product in this table, the necessary input data and methods used for their computation are specified alongside their associated uncertainty percentages and the studies from which these products are sourced. In general, spherical particles are assumed for both water droplets and ice particles. For droplets, a log-normal distribution is presumed, whereas empirical relations are utilized for ice particles.

Table 2. Cloudnet microphysics products used in this study, their inputs and assumptions

Cloudnet products	**Input role**	σ_{re}/r_e	**References**
Liquid water content (LWC)	Z and β: cloub base and top identification T, P, assuming an adiabatic lifting MWR or CDR LWP to scale LWC	15%	Frisch et al. (1998)
Ice water content (IWC)	Z and T profiles	-35% to 90%	Robin J. Hogan et al. Julien Delanoë et al.
Droplet effetive radius (r_{liq})	Z, and N and σ of lognormal PSD	15%	S. Frisch et al.
Ice effective radius (r_{ice})	Z and T	50%	Hannes Griesche et al.

The liquid water content (LWC) retrieval assumes a log-normal distribution for warm cloud droplets lifting adiabatically from its base. Its relation dependent on atmospheric temperature (T) and pressure (P). This method incurs a relative uncertainty of 15% as showed by A. S. Frisch et al.. At our station, T and P values utilized by the LWC retrieval are provided by the

European Centre for Medium-Range Weather Forecasts (ECMWF). The retrieval equation is applied exclusively to liquid pixels within cloud boundaries, which are determined by Doppler radar reflectivity (Z) and ceilometer-lidar attenuated backscatter coefficient. The liquid pixels, as ice, insect and aerosol pixels are given by the cloudnet target classification product, discussed in section **??**. Additionally, LWC values are scaled to align with the Liquid Water Path (LWP) derived from the microwave radiometer. The liquid cloud droplet effective radius (r_{liq}) is retrieved by assuming a log-normal size distribution for warm cloud droplets. This method is insensitive to droplet number concentration and the logarithmic spread of the distribution (S. Frisch et al.). Consequently, its primary source of uncertainty is the reflectivity factor error, resulting in a relative error of 15% for r_{liq} retrievals. This relationship is applied exclusively to liquid pixels.

The ice water content (IWC) retrieval employs an empirical equation as determined by Robin J. Hogan et al., known as the Z-IWC-T relation, which depends on radar reflectivity and temperature. Compared to other microphysical variables, IWC retrievals has the largest relative uncertainties. For the W-band, the relative error is +55% to -35% for temperatures within the range of -20°C to -10°C, and +90% to -47% for T < -40°C. While temperature is a significant parameter in IWC error estimation, Pablo Saavedra Garfias et al. demonstrated that the primary source of uncertainty is the reflectivity factor error. A typical reflectivity error of 3% can result in a 12% error in the ice water path (IWP), which is the integration of IWC over the atmospheric column. Therefore, errors in IWC can be relatively large. The Z-IWC-T relation is applied only to ice pixels within cloud boundaries. Next, the ice cloud particle effective radius (r_{ice}) is obtained following Julien Delanoë et al., who assumed the same empirical relation for IWC and the visible extinction coefficient (α) derived by Robin J. Hogan et al.. This approach, termed the Z-T relation, depends on both T and Z, and has aThis approach, termed the Z-T relation, depends on both T and Z, and has a relative uncertainty of 50% of r_{ice} (Hannes Griesche et al.). In generall, this high relative errors emphasize how challenge is retrieving ice properties accurately.

3 Methods

3.1 Cloud assessment by hydrometeor grouping

Many recent studies have been employed the algorithm of cloud classification by time (or by profile) with the aim of classify clouds (Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.) and analyze their statistics. However, this algorithm admit different cloud typing inside the same cloud as a whole, assigning similar physical properties for different clouds. In the appendix A, we describe this algorithm, adapting the criteria used by Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.. In another way, this section proposes a different classification algorithm, based on connecting groups of near hydrometeor.

First, the algorithm identify all hydrometeor pixels in the cloudnet target classification to generate a cloud mask. In sequence, hydrometeors with only two pixels of separation (at any direction) were assumed to be close enough to be considered part of the same cloud. Hence, a method of pixel dilatation and pixel contraction were applied to link these groups of closed hydrometeors. This method is explained in more details in appendix B. Once the groups in target classification were identified, a sequence of verification is applied. First, the total number hydrometeor (without considering rain such as Drizzle or rain pixels) inside each group must be greater than 100 pixels to be considered as cloud. Otherwise, it was deemed as noise, being of unknown type.

For each cloud group, the percentage of each hydrometeor is calculated in order to apply the following classification scheme (shown in figure 3): if a cloud comprised more than 70% of Droplets and drizzle plus Droplets, it was classified as a liquid cloud (Liquid criteria). If it contained at least 10 connected rain pixels, it was further classified as a precipitating liquid cloud (Rainy criteria); otherwise, it remained classified as a non-precipitating liquid cloud. Similarly, if the previous verification was not satisfied, if this group contain more than 90% of ice or less than 10% of Droplets plus Ice & droplets, then it is classified as ice cloud (Ice criteria). Next, the same rain verification as in the liquid case is done to classify this group as purely ice cloud or precipitating ice clouds. Otherwise, cloud is classified as mixed-phase or precipitating mixed-phase through the same rain verification as other clouds. In relation of precipitating ice and precipitating mixed-phase clouds, an extensive analysis of individual case studies revealed that Drizzle & droplets only appears below the melting layer together with Drizzle & rain droplets. Therefore, to avoid an underestimation of cloud base height due to drizzle, we removed these pixels for these clouds since they are probably rain.

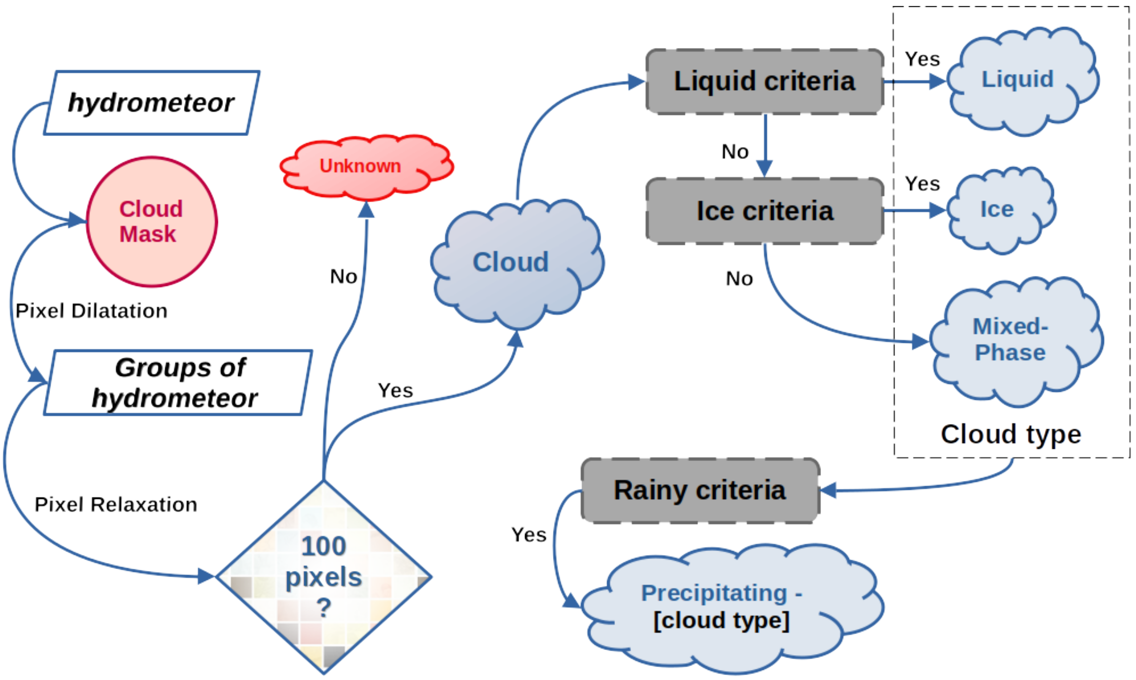


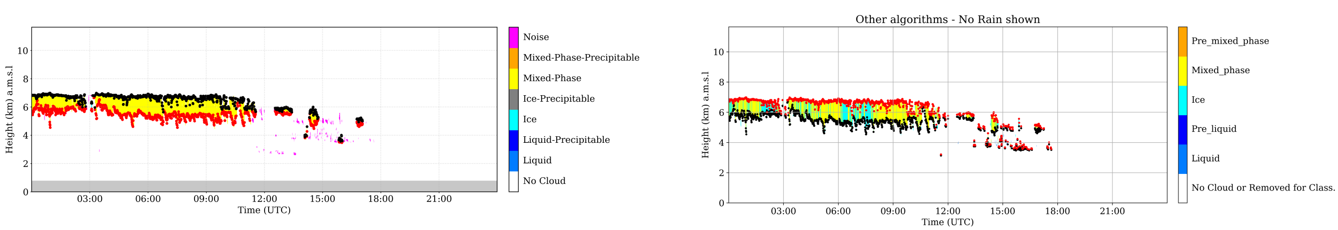
Figure 3. Scheme of the cloud classification algorithm by hydrometeor grouping

After the cloud classification, each cloud is then classified as single or multi-layer over time. Initially, time intervals where cloud overlap are identified. Overleaped pieces of clouds were classified as multi-layer, whereas non-overlapping pieces in principle can be classified as single layer. We then analysed cloud profiles for each cloud individually, acknowledging that a clouds can exhibit multiple layers with itself. If a cloud had a second layer separated by only one pixel at some time interval, that interval is identified as containing multi-layer. Otherwise, the clouds at that interval is classified as single layer. For the single layer case, cloud properties were calculated as follows: the cloud base and top height is determined by the first and last height pixels within the cloud layer, respectively. Cloud thickness is obtained by the difference between cloud base and top height pixels.

Taking into account ice, melting and rain hydrometeor misclassification around mid-day as discussed in section 2.1.2, which can result into to low level ice and precipitating ice clouds misclassification, these cases were filtered. Basically, all these clouds which were identified with mean cloud base below 4 km (above mean sea level) and average cloud thickness below 700 meters were reclassified as noise. These values were determined empirically by studying many cases separately. Finally, for those cases with no clouds, time interval is classified as clear sky condition.

3.2 Cloud assessment

A comparative analysis between the two classification algorithms were conducted in order to evaluated their applicability for further analysis. Here, we aim to provide insights into the correlation on cloud statistics between different cloud categories generated by these methods. The findings from this comparison will inform the selection of the most appropriate method for future atmospheric studies and cloud analysis. In this framework, Figure 4 shows a case study at 6 May of 2023 in Granada comparing cloud classification by time and by hydrometeor grouping for an mixed phase cloud, where red and black dots represent the cloud base and top, respectively. As expected, the classification by time on the right panel lead to the fragmentation of clouds into different types. In this case, the same cloud are being classified as ice and mixed-phase by time. On the other hand, in the left panel, the classification by grouping preserves the homogeneity of clouds and classify it as mixed-phase only, resulting in a more realistic classification.



(a) Cloud classification using the new method.

(b) Cloud classification using the old method.

Figure 4. Temporary figure (needs target classification). Comparison of cloud classification methods. The left panel illustrates the new method, which preserves cloud homogeneity and provides a more realistic classification. The right panel shows the old method used in recent studies, where cloud classification is time-based.

As a consequence, the time-based method of cloud classification can generate high correlations between cloud categories in terms of frequency of occurrence and cloud properties. In relation to the cloud occurrence, right panel of Figure 4 suggest that both ice and mixed phase clouds will have similar daily cloud occurrence and average cloud base. However, this is occurring because of the method of classification, which could affect further cloud statistic. To prove that, the 6-year dataset at the Granada station was used to compute the daily average IWP for ice and mixed-phase clouds using the classification by time. Figure 5 shows the closeness daily ice amount between these two clouds (mixed phase in above panel and ice cloud in the bottom panel) and the similar pattern suggest a correlation generated by the cloud classification. It is worthy to note that this section is not focused on the analysis of IWP, this will be done in section XXX. Instead of this, here we centered on the correlation generated by the classification algorithms. Therefore, it was clearly observed that high correlation among cloud categories can generated similar physical properties among than, resulting in a misinterpretation of the results. It proceed because its difficult to attribute similar properties between clouds (formed by different mechanisms) as a realistic phenomenon that happens in the nature or attribute to the method of classification.

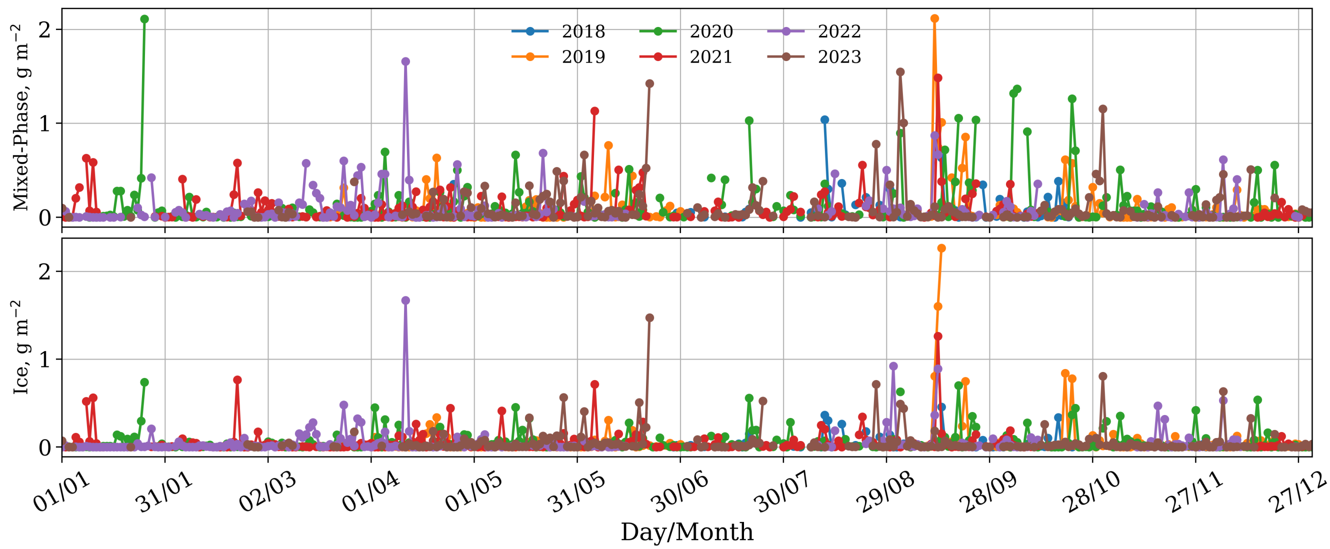


Figure 5. Time series of daily Mixed-Phase (top) and Ice (bottom) cloud ice water path from 2018 to 2023. Each year is represented by a different colored line.

Therefore, in order to quantify the correlation in between cloud categories generated by these two algorithms of classification, we calculate the daily cloud occurrence to access the correlation between cloud types. Figure 6 shows the pearson correlation coefficients of daily cloud occurrence between cloud types for the classification by hydrometor grouping on the left, and for the classification by time on the right. As can be seen, the new proposed method demonstrate generally weaker correlations among cloud types, with most values clustering closer to zero. For example, the highest correlations observed were 0.19 between Liquid and Liquid-Precipitable clouds and 0.17 between Liquid and Mixed-Phase-Precipitable clouds, indicating relatively weak relationships. Additionally, some negligible negative correlations were observed, such as between Ice and Mixed-Phase-Precipitable clouds (-0.1). This suggests that the new method may provide a clearer distinction between cloud types, which is advantageous for analyses requiring precise classification. In addition, computed cloud properties will be not dependent on the method as they must be, avoiding some misinterpretation of the results.

In contrast, the classification by time exhibited stronger and more pronounced correlations. Notably, the correlation between Ice and Mixed-Phase clouds was 0.54, and between Liquid and Mixed-Phase clouds was 0.44, highlighting significant inter-dependencies between these cloud types. Furthermore, the time-based method showed higher correlations between Liquid and

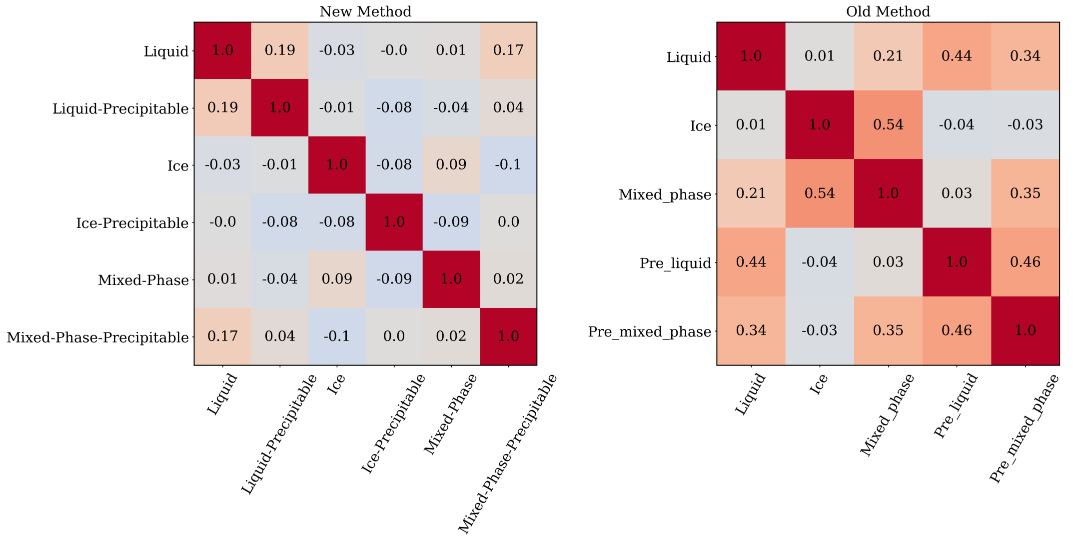


Figure 6. Correlation matrices comparing the daily occurrence between different cloud types using two methods. In the left, the New Method exhibits generally weaker correlations, indicating clearer distinctions between cloud types. In the right, The Old Method shows stronger correlations, suggesting more pronounced interdependence between cloud types

Liquid-Precipitable (0.34), Ice and Mixed-Phase-Precipitable (0.35), and Liquid-Precipitable and Mixed-Phase-Precipitable (0.46), indicating a less distinct separation between cloud types. These higher correlation values are a consequence of the classification scheme defined in this method. The cloud classification by time can fragment them, allowing tagging cloud profiles in different cloud categories, while they still inside the same cloud structure. Overall, the comparison indicates that the new method, with its generally weaker correlations, may be better suited for distinguishing different cloud types. These findings underscore the importance of selecting an appropriate analytical method to study cloud properties between different cloud types.

Appendix A: Cloud assessment by time

This method have been used by recent studies such as Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.. Each study has adopted some different criteria to classify single layer clouds, but they all classify cloud layers by profile. Here, we adopted some criteria based on these studies and employed the following method: cloud layers in atmospheric column were classified trough pixel sequence inquiry. These pixels sequences were divided in the following 5 groups according to Răzvan Pîrloagă et al.: liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only "ice" pixel sequence was encountered. As well as, mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified whether at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Different from previous studies, whether rain pixels are found with more than 20 pixel (256-1022 m) of distance from cloud base, the cloud are not classified as precipitable, assuming that rain droplets are not falling from these cloud layer. In all cases, for being considered a cloud layer, a minimum sequence of 3 cloud pixels (38-153 m) (**?**) must be satisfied, in addition to a minimum sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" to separate multi-layers. Therefore, free hydrometeor layers in the neighbourhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on **??**, but theses values are not currently well defined, their determination has been guided by empirical experience.

Appendix B: Pixel dilatation

The dilation method consist in expand the ones in the cloud mask to connect the closest groups of hydrometeors, which are likely to belong to the same cloud. The dilatation follows the scheme bellow, where hydrometeor pixels are represented in blue. Left image represents the original hydrometeor pixels where two groups of clouds were identified. Thus, a dilatation of 1 pixel is applied in all directions, represented by the red pixels in the right image. Therefore, the dilatation process will unite these groups with at least 2 pixels of separation distance. Since they are connected, the cloud classification algorithm will save the coordinates of these cloud pixels in just one group (not two, as previously identified).

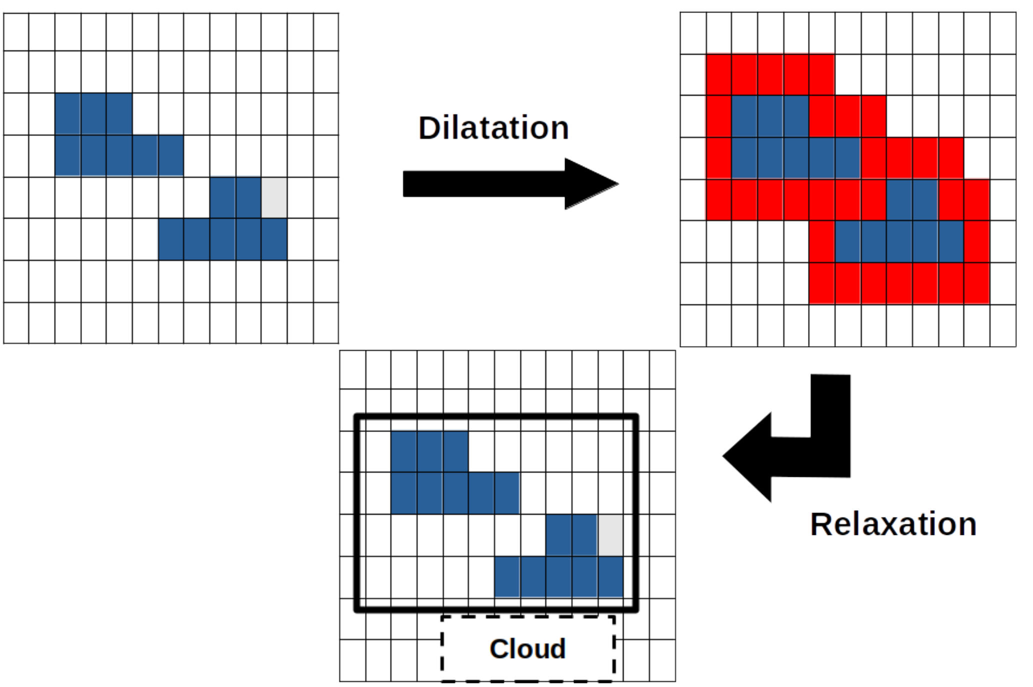


Figure B1. Pixel dilatation and pixel contraction illustration. Left panel shows two hydrometeor groups. Right panel still showing these two clouds and the dilated pixels in red. Since these clouds are connected, the classification algorithm will understand there is only one cloud. Bottom panel shows the result of the pixel contraction, returning cloud pixels to its original position

Next, a contraction process is done to return the coordinates of true hydrometeors to the original position. The contraction will remove the coordinates of fake hydrometeor pixels (pixels in red on the above figure). Afterwards, the classification algorithm can follows its chain and calculate the percentage of each hydrometeor type inside each group to classify different clouds in the cloudnet product.

Acknowledgements. TEXT

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